

Lab 11 - Debugging Report

App: Taskly | Student: Tran Nguyen Khanh (SE172279)

Tool: Flutter DevTools | Device: Medium Phone API 35 emulator

1. Goal

Use Flutter DevTools to inspect the Taskly UI widget tree and capture a short Performance session while interacting with the task list (add / toggle / navigate).

2. Setup

- Run: flutter run (debug mode on Android emulator).
- Open DevTools at `http://127.0.0.1:9100`
- Inspector tab for widget tree; Performance tab for frame timeline + rebuild stats.

3. Widget Inspector findings

Screenshot: `evidence/widget_tree.png` (also root: `devtools_screenshot_widget_tree.png`)

- Tree path: MyApp > MaterialApp > TaskListScreen > Scaffold > Column.
- Input row: Padding > Row > TextField + ElevatedButton (label Text "Add").
- List: Expanded > ListView > Dismissible > ListTile (Checkbox + title/subtitle Text).
- Selected widget Text("Add") size ~25.4 x 20.0 with vertical padding 10.0.
- Emulator state matches tree: tasks AAA/BBB (checked), CCC (unchecked).

4. Performance findings

Screenshot: `evidence/performance_trace.png` (also root: `devtools_screenshot_performance.png`)

- Engine: Impeller; Flutter frames chart shows UI (pink) and Raster (blue) time.
- Some frames show jank (above ~16ms / 33ms), common in debug mode on emulator.
- Rebuild Stats: TaskListScreen 2, ListTile 6, Checkbox 6, ElevatedButton 4, TaskDetailScreen 2.
- Hot spots map to `task_list_screen.dart` and `task_detail_screen.dart`.

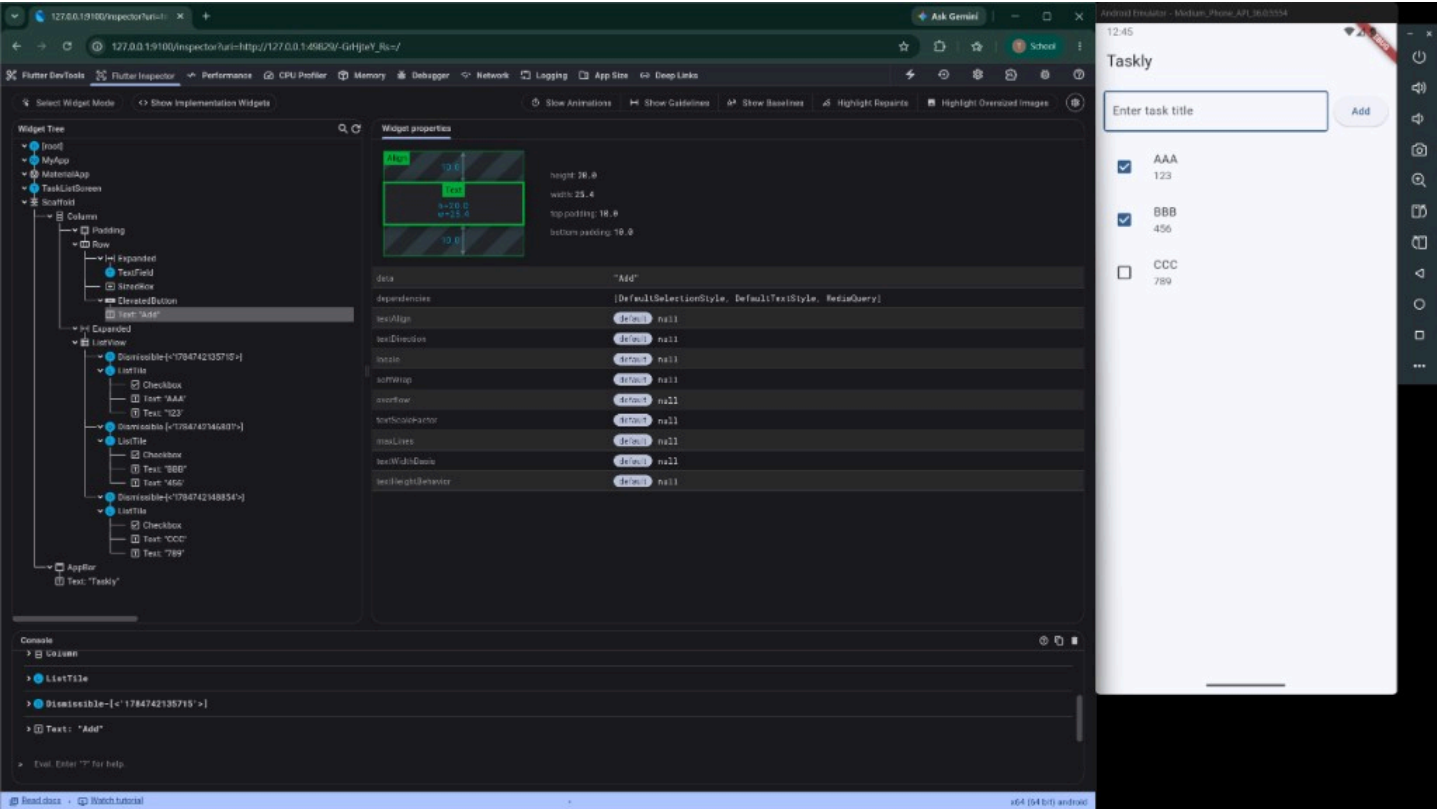
5. Issues observed & notes

- No functional UI bugs found; list, checkboxes, and Add control render correctly.
- Jank frames are expected in DEBUG; use profile/release for real perf tuning.
- Rebuild counts for ListTile/Checkbox match list interactions (add/toggle/detail).

6. Conclusion

DevTools confirms the Taskly widget hierarchy and interaction rebuilds. Screenshots in the appendices are the required evidence for Lab 11 Part B.

Appendix A - Widget Inspector screenshot



Appendix B - Performance screenshot

